RESEARCH REPORT · JACOBIAN LENS

Representational readout and causal transport in language models

Generated 2026-07-19 from recorded JSON artifacts.

Abstract. We reproduced the local Jacobian-lens readout path and extended it into controlled causal swap experiments across decoder models from 135M to 27B parameters. The model-matched 27B JLens produced stronger target-rank movement than logit-lens and random matched controls in the current verbal-report, two-hop, and flexible-generalization runs. Existing tuned results are explicitly labelled `tuned-wiki-small-v0`: model-matched Wikitext pilots, not a strong reproduction.

Current status: model-matched `tuned-pile-repro-v1` predictive and causal results are complete for SmolLM2-135M, Qwen3-0.6B, and Qwen3.5-4B. Qwen3-1.7B fitting is complete and its canonical 16.4M-token held-out gate is still running; its short-2M causal suite remains explicitly provisional. No provisional result is pooled into the validated Pile causal table.

Contents

Model and intervention coverage

Model catalog

This catalog describes the subjects, not the interventions. “Base/pretrained” means the checkpoint name does not identify an Instruct variant; it is not a claim that the model lacks reasoning behavior. Precision describes the recorded run, and can differ between fit, evaluation, and causal inference.

Subject	Exact checkpoint ID	Parameters	Family	Checkpoint flavor	Architecture	Recorded run precision
SmolLM2-135M	`HuggingFaceTB/SmolLM2-135M-Instruct`	135M	SmolLM2	Instruction-tuned	Decoder-only Transformer	BF16 compute; unquantized base on RTX 3090
Qwen3-0.6B	`Qwen/Qwen3-0.6B`	0.6B	Qwen3	Base/pretrained checkpoint	Decoder-only Transformer	BF16 compute; unquantized base on RTX 3090
Qwen3-1.7B	`Qwen/Qwen3-1.7B`	1.7B	Qwen3	Base/pretrained checkpoint	Decoder-only Transformer	BF16 compute; unquantized base on RTX 3090
Qwen3.5-0.8B	`Qwen/Qwen3.5-0.8B`	0.8B	Qwen3.5	Base/pretrained checkpoint	Decoder-only Transformer	BF16 compute; unquantized local fit
Qwen3.5-4B	`Qwen/Qwen3.5-4B`	4B	Qwen3.5	Base/pretrained checkpoint	Decoder-only Transformer	BF16 compute; unquantized local fit
Qwen3.6-27B	`Qwen/Qwen3.6-27B`	27B	Qwen3.6 / Qwen3.5-family	Base/pretrained checkpoint	Hybrid GDN/attention decoder	BF16 compute; NF4 4-bit frozen base for the Modal tuned fit and 27B causal runs

Intervention definitions

All four methods use the same causal protocol: choose a source and target concept, construct their residual-stream directions at the selected layer, swap the two coordinates with a two-vector pseudoinverse frame, and measure the target answer's rank after the model continues forward. For the 27B runs, the intervention layers are 16, 32, and 60, and the patch is applied at every prompt position.

Method	Direction used	What is learned?	Control/intervention meaning
J-lens	J_{layer}^T times the model's unembedding row for the token	A model- and layer-specific Jacobian lens fitted from prompts	Swap coordinates in the learned global-workspace/readout basis
Logit lens	The raw unembedding row for the token; equivalent to J = I	Nothing	Same coordinate-swap machinery, using ordinary logit directions
`tuned-wiki-small-v0`	$(I + W^T)$ times the unembedding row, where W is the learned affine translator	Model- and layer-specific affine translators trained by KL-to-final-logits on short Wikitext-2 fits	Current tuned causal rows; exported translator-basis swap, with nonlinear final RMSNorm not folded into this patch
`tuned-pile-repro-v1`	$(I + W^T)$ times the unembedding row, where W is the learned affine translator	Same translator objective, fit on Pile validation and accepted only after held-out Pile-test validation	Validated causal rows pending; Qwen3-1.7B has a separately labelled short-2M provisional track
Random matched	A random residual direction scaled to the norm of the corresponding coordinate-swap delta	Nothing	Controls for generic perturbation magnitude rather than semantic direction

How to read the random control: it is not expected to have zero successes. A norm-matched random perturbation can move a target token up or down by chance; its success rate is the measured noise floor for this rank-based metric. A method should therefore be judged by its excess over random, not by raw improvement alone. If random approaches the semantic method, the intervention may be exploiting generic residual sensitivity rather than the intended concept direction.

Tuned-lens training-data scale

Variant	Corpus	Fit material	Optimization	Held-out evaluation	Status
`tuned-wiki-small-v0`	Wikitext-2 raw train	32,768 sampled tokens for the smaller models; 65,536 for the 27B pilot	100 steps, 128-token chunks, learning rate 1e-3	Not part of the original pilot fit	Used in current causal plots; sanity-check only

Variant	Corpus	Fit material	Optimization	Held-out evaluation	Status
`tuned-pile-repro-v1`	The Pile validation split	16,384 chunks per model; 2,097,152 sampled training tokens	4,096 steps, 128-token chunks, learning rate 1e-3	Pile test, target 16.4M tokens	Full predictive/causal track complete for SmolLM2, Qwen3-0.6B, Qwen3.5-0.8B, and Qwen3.5-4B; Qwen3-1.7B full gate in progress

These are materially different training regimes. The Pile first pass exposes the translators to roughly 32–64 times more token positions than the Wikitext pilots, and it has a genuinely held-out test evaluation. Therefore a future Pile causal result must be compared against the corresponding model's Pile-trained lens, not silently pooled with the Wikitext rows.

The 27B released J-lens is a separate model-matched 1,000-prompt artifact. The new 27B tuned lens was fit separately using 512 Wikitext chunks of length 128 for 100 optimization steps and is labelled `tuned-wiki-small-v0`. The stronger `tuned-pile-repro-v1` track is now being staged, starting with Qwen3-0.6B. Logit and random are not trained lenses. Thus the causal comparison holds the model, prompt fixture, patch positions, layers, and swap rule fixed while changing the direction basis.

Measurements and estimands

The plots repeat the key labels, but the estimands are defined here once so the reader can interpret every task consistently. Unless noted otherwise, each scored observation is one valid prompt/item at one intervention layer; multiple layers from the same prompt are retained together for uncertainty estimation.

Measurement	Definition	Interpretation
n	Number of scored item–layer observations after tokenization and validity filtering.	Denominator for the task's aggregate rate; it is not always the number of unique prompts.
Improved conditions (%)	100 × fraction where target rank before − target rank after is positive.	How often the intervention moves the intended target upward; positive does not require top-1.
Δrank	Target rank before − target rank after.	Positive is improvement; the median is the primary robust effect-size summary, while the mean is more sensitive to outliers.
Target top-1 / top-5 / top-10	Fraction of post-intervention observations where the intended target has the indicated rank threshold.	Absolute endpoint performance after intervention, distinct from rank movement.

Measurement	Definition	Interpretation
Excess over random	Method improvement rate minus the matched-random improvement rate on the same task/model.	Estimated direction-specific effect above generic norm-matched perturbation sensitivity.
95% bootstrap interval	Cluster bootstrap over prompts/items, keeping their scored layers together.	Uncertainty over examples, not an assumption that layers are independent samples.

The verbal-report task uses target top-10; the two-hop and flexible tasks use target top-5. Random matched controls are not expected to have zero improved conditions: they establish the noise floor for this rank-based metric.

Larger ProofWriter confirmation

We selected readout layers on a source-disjoint 300-item set, locked them, and then evaluated a separate 1,500-item confirmation set. Both sets contain 100/500 examples per class, respectively, and no underlying ProofWriter theory appears in both sets. Intervals resample theory groups rather than treating related questions as independent.

Model	Readout	Accuracy	95% clustered CI	Macro-F1	Unknown recall
SmolLM2-135M	final	48.5%	[46.0%, 51.0%]	0.385	0.0%
SmolLM2-135M	jlens	58.5%	[56.0%, 61.0%]	0.488	3.2%
SmolLM2-135M	logit	41.8%	[39.3%, 44.3%]	0.328	0.0%
Qwen3-0.6B	final	39.1%	[36.7%, 41.6%]	0.295	31.0%
Qwen3-0.6B	jlens	42.9%	[40.3%, 45.4%]	0.393	65.6%
Qwen3-0.6B	logit	37.3%	[34.8%, 39.7%]	0.298	60.8%
Qwen3-1.7B (1,500)	final	35.1%	[32.7%, 37.5%]	0.208	6.0%
Qwen3-1.7B (1,500)	jlens	55.1%	[52.6%, 57.5%]	0.556	55.2%
Qwen3-1.7B (1,500)	logit	49.0%	[46.5%, 51.5%]	0.452	62.0%
Qwen3-1.7B (4,800)	final	35.9%	[34.6%, 37.3%]	0.222	8.8%
Qwen3-1.7B (4,800)	jlens	57.3%	[55.9%, 58.7%]	0.579	61.8%
Qwen3-1.7B (4,800)	logit	51.2%	[49.8%, 52.6%]	0.462	71.9%
Qwen3.5-4B (4,800)	final	59.3%	[57.9%, 60.7%]	0.533	14.2%
Qwen3.5-4B (4,800)	jlens	67.5%	[66.1%, 68.8%]	0.665	43.4%

Model	Readout	Accuracy	95% clustered CI	Macro-F1	Unknown recall
Qwen3.5-4B (4,800)	logit	57.4%	[56.0%, 58.8%]	0.552	58.1%

Supervised linear-probe comparison

A multinomial logistic-regression probe was fitted separately at each hidden layer on the 300-item labeled selection set; the best layer was locked before evaluation on the 4,800-item source-disjoint confirmation set. This is a task-specific upper comparison, not a task-agnostic lens.

Qwen3-1.7B

Selected layer	Accuracy	Macro-F1	Unknown recall
layer 10	64.0%	0.636	46.8%

Qwen3.5-4B

Selected layer	Accuracy	Macro-F1	Unknown recall
layer 29	80.2%	0.803	75.3%

Task-specific J-lens pilot

Starting from the generic layer-21 J-lens, we learned a rank-16 low-rank update on the 300 labeled selection items and evaluated on the same 4,800-item confirmation set. This is a separate supervised variant, not the standard task-agnostic J-lens.

Readout	Accuracy	Macro-F1	Unknown recall
Generic J-lens	57.3%	0.579	61.8%
Label-adapted J-lens (rank 16)	66.5%	0.625	25.6%
Same-layer linear probe	72.5%	0.723	58.2%

This is an observational three-way label readout, not a causal swap test. J-lens is task-agnostic; the probe comparison below is supervised and uses labeled selection examples. Neither result establishes universal superiority over final-layer generation.

Are two-hop reasoning and flexible generalization actually distinct?

At the level of the current fixtures, the distinction is weaker than the task names suggest. Both intervene on an argument-like representation and ask whether a downstream token changes appropriately. The two-hop prompt adds a nested relation; the flexible prompt makes the relation look like a named function. That may be a meaningful compositional difference, but it is not yet a strong measurement separation.

The correlation table uses the six shared models and aggregate improved-condition rates, so it is descriptive and underpowered rather than a formal validation of task independence. A high positive correlation means the tasks may be tracking a shared argument-substitution sensitivity; it does not prove that they are identical.

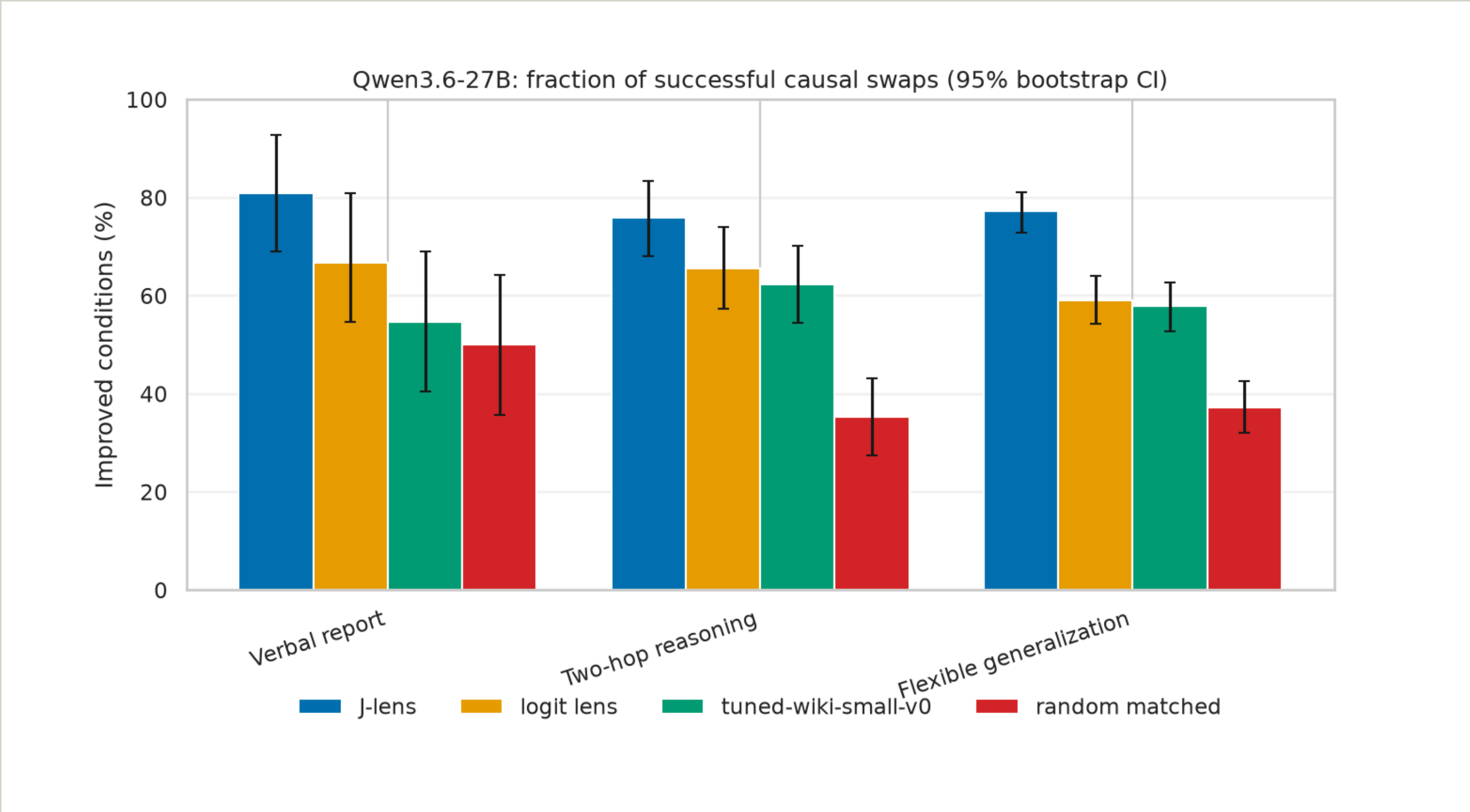
Current fixture	Swap	Expected shift	Operational interpretation
The capital of France is the city of	France → Canada	Paris → Ottawa	Recompute a one-argument function after an entity substitution; current flexible-generalization fixture.
Fact: The language spoken in the country where the Amazon River ends is	Brazil → Mexico	Portuguese → Spanish	Carry an intermediate entity through a nested relation; current two-hop fixture.

> **Interpretive consequence:** we should not present the current two-hop and flexible suites as independent evidence without qualification. The next stronger version should hold the surface operation constant while varying whether the answer requires one substituted argument or a genuinely necessary intermediate chain, and should include matched controls that distinguish direct lookup from composition.

Qwen3.6-27B headline results

Verbal report task: replace one entity in a factual prompt and test whether the model's next answer or continuation shifts toward the substituted entity. The headline plot summarizes the fraction of causal swap conditions where the target answer's rank improved.

Figure: 27B headline improvement rates [#fig-27b-improvement-rate]



27B bootstrap uncertainty

Intervals use a deterministic cluster bootstrap over prompts/items, resampling each prompt as a unit while retaining its multiple scored layers together. This is more conservative than treating every layer as independent.

Method versus random control

Differences are paired success-rate contrasts in percentage points: JLens or tuned lens minus the random matched control. A positive interval entirely above zero is the clearest evidence in these pilot metrics that the method beats random for that model and experiment.

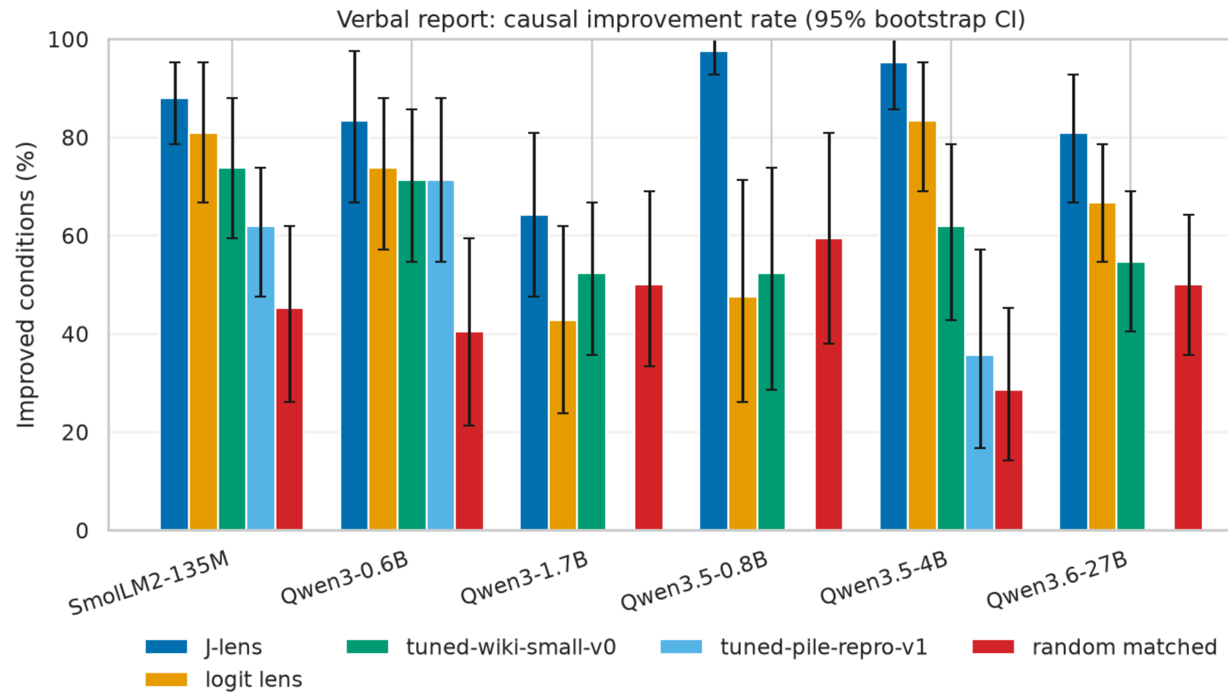
Cross-model causal comparisons

These plots show the fraction of scored conditions in which the target improved after the intervention. Wikitext and full Pile tuned lenses are displayed as separate series; the earlier Qwen3-1.7B short-2M diagnostic is excluded until its full held-out Pile protocol is complete. Prompt counts and tokenization skips should still be inspected in the per-experiment logs.

Verbal report

Task: ask the model to produce a category member, then intervene on the representation of the selected source concept and test whether the target candidate rises at the answer position.

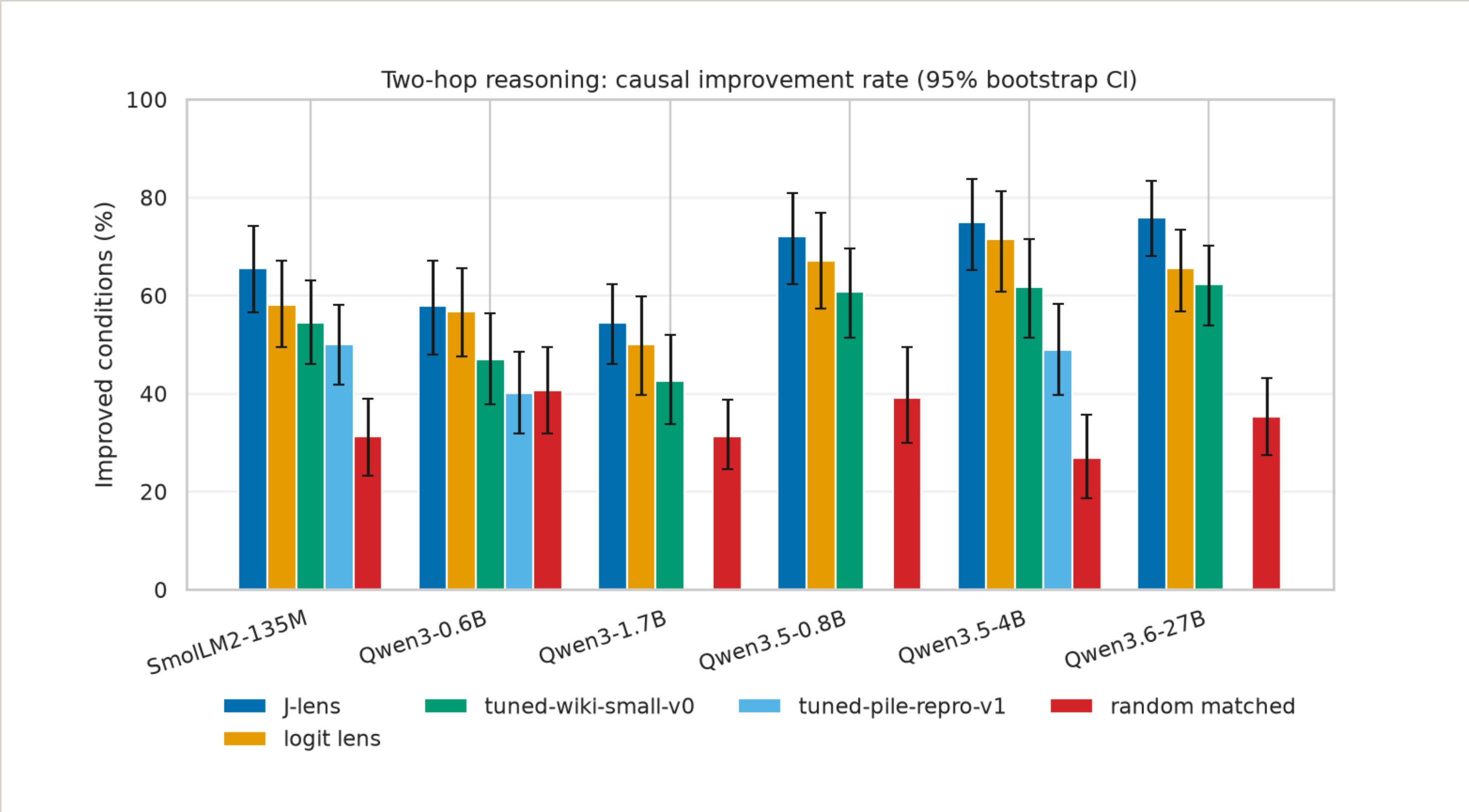
Figure: Verbal-report cross-model comparison [#fig-verbal-cross-model]



Two-hop reasoning

Task: the prompt states two linked facts; the intervention swaps the representation of the intermediate entity, and success means the model's downstream answer moves toward the answer implied by that substituted intermediate.

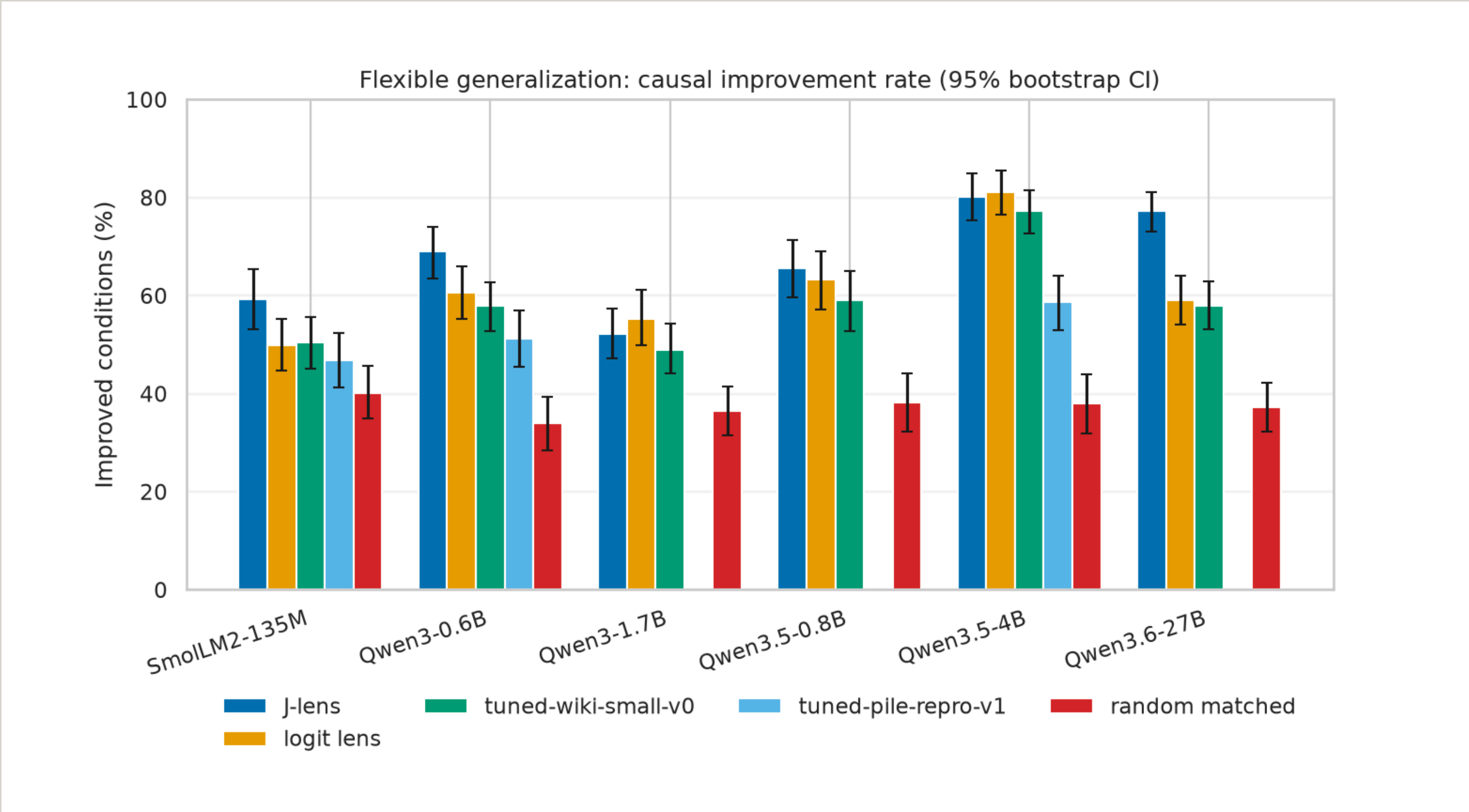
Figure: Two-hop reasoning cross-model comparison [#fig-multihop-cross-model]



Flexible generalization

Task: substitute one argument for another, then test whether the model correctly carries that replacement through several downstream functions such as ordering, comparison, arithmetic, or relational use.

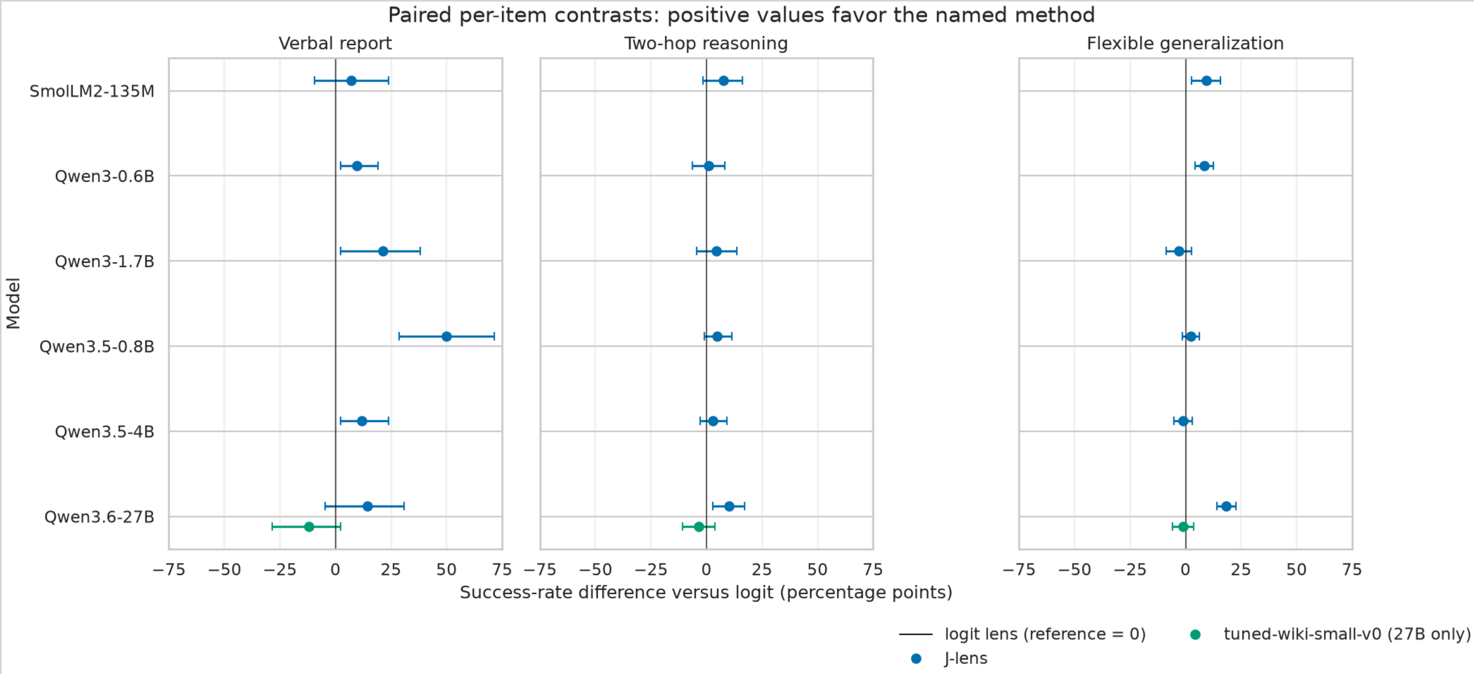
Figure: Flexible-generalization cross-model comparison [#fig-flexible-cross-model]



Paired per-item method contrasts

Aggregate bars answer how often each method succeeds overall; they do not show whether the same items improve under one method and fail under another. This analysis pairs methods on the same item and averages over that item's scored layers, then reports the difference in success rate, a prompt/item bootstrap interval, and a paired sign-flip p-value. Positive values favor the named method over logit lens. These are paired descriptive/inferential summaries, not independent-sample tests, and each saved artifact remains a separate row rather than being pooled across models or tasks.

Figure: Paired per-item contrasts versus logit lens [#fig-paired-causal-contrasts]



> **Quick read:** logit lens is the reference method and therefore appears as the vertical zero line, not as a separate point series. Points to the right mean the named method wins more same-item comparisons than logit lens; intervals crossing zero are inconclusive at this uncertainty level. The green tuned-lens point is 27B-only because no other model currently has a paired tuned-versus-logit artifact. The plot is most useful for seeing whether an apparent aggregate advantage is consistent across models and tasks, rather than driven by one large bar.

The corresponding machine-readable artifact is `data/analysis/paired_causal_effects.json`. Random-versus-logit controls remain in the 27B contrast table and the underlying artifact because their role is to establish the perturbation noise floor, not to replace the method-versus-method comparison.

Effectiveness versus model size

The first view uses the intuitive absolute improvement rate. The second subtracts the matched random-control improvement rate, which is a better summary of technique-specific effect. The third shows median rank movement, capturing effect magnitude rather than only whether movement was positive. All use parameter count on a logarithmic x-axis and item-cluster bootstrap intervals.

Figure: Absolute effectiveness versus model size [#fig-parameter-effectiveness]

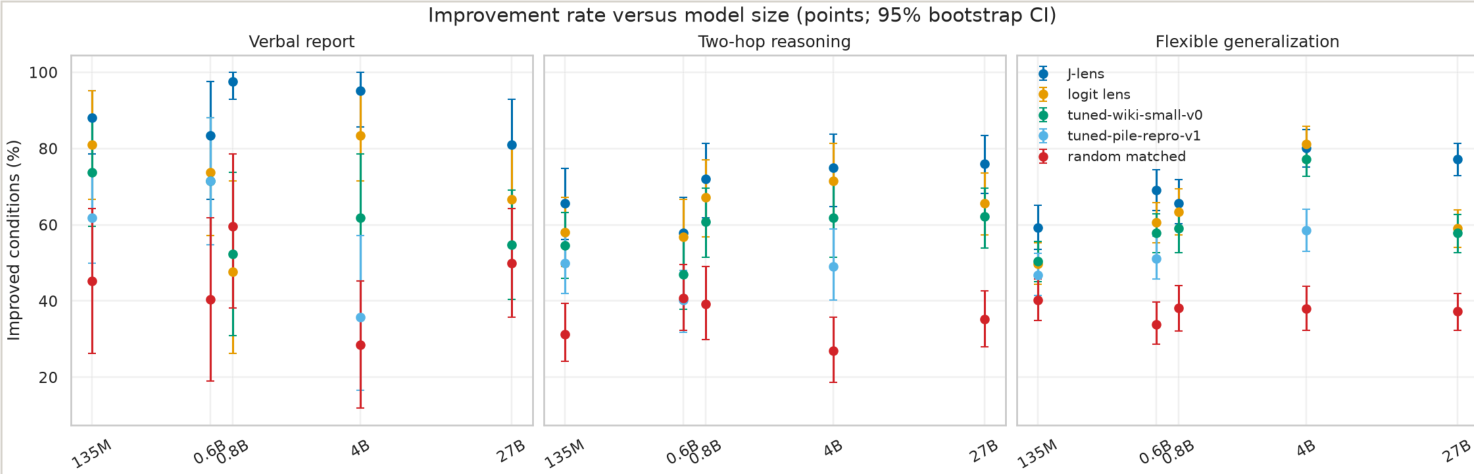


Figure: Excess over random versus model size [#fig-parameter-excess-random]

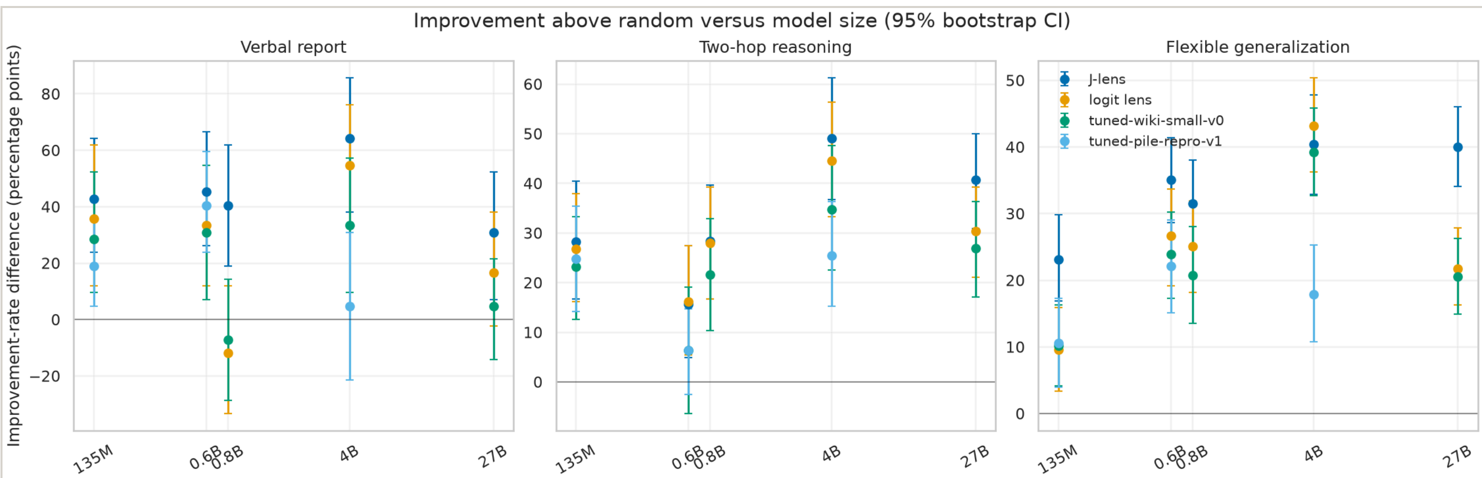
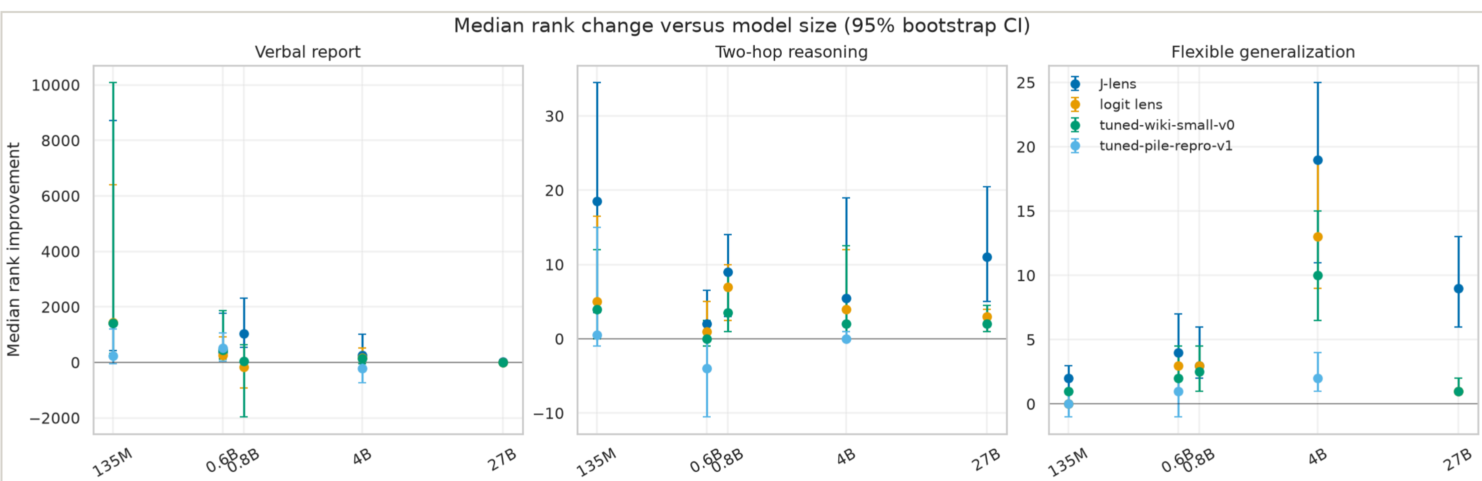


Figure: Median rank movement versus model size [#fig-parameter-median-delta]



27B creation-cost evidence

The primary efficiency question is now stated at the target model scale: what did it take to create each 27B artifact? This is separate from causal coverage. The causal plots include the released model-matched J-lens artifact, logit lens, random matched controls, and the fitted Wikitext tuned lens; only the tuned Wikitext row currently has our 27B fit telemetry. Logit and random have no learned artifact by design, and the J-lens creation run was not recorded in our checkout. The raw times are therefore evidence, not a causal ranking. The method-native rates are also not interchangeable: J-lens uses prompt-layer evaluations, tuned lens uses optimizer steps, and logit lens has no fit. The broad model-scale plot is retained below as exploratory context only.

Memory evidence by intervention

Peak VRAM in GiB is a directly comparable resource measurement only when model, precision, batch size, and workload are specified. A missing value is not zero. Logit lens has zero learned-artifact creation cost, but it still consumes the base model's runtime memory.

Held-out Pile predictive validation

These are predictive checks for the tuned-lens training track, not causal swap results. Each row is evaluated on the held-out Pile test stream; lower KL to the frozen final model distribution is better. The three KL columns are first, middle, and final transformer layers.

Validated Pile causal suites

These rows are kept separate from the Wikitext-pilot causal plots. They appear only after the corresponding held-out predictive artifact passes validation, and report the Pile-trained tuned basis alongside logit and random controls.

The earlier Qwen3-1.7B short-2M diagnostic artifacts are retained in the data directory for provenance, but are intentionally excluded from comparative plots and result tables. The full model-matched Pile protocol is being completed instead.

Interpretation and open questions

- The original readout path runs locally, and the 27B PyTorch path works in guarded 4-bit NF4 mode.
- The released 27B JLens is model-matched, has a 1,000-prompt fit, and supports the current causal runs.
- JLens, logit, and random controls have been run on the three main causal protocols.
- `tuned-wiki-small-v0` comparisons exist for the smaller models and 27B; their results are pilot evidence only because the fits used short Wikitext runs.
- The proper Pile validation/test reproduction is being staged as `tuned-pile-repro-v1`, starting with Qwen3-0.6B and proceeding upward only after held-out predictive checks pass.
- A model-matched Qwen3.6-27B Wikitext pilot is available. Its causal comparison uses its exported affine translator basis; the nonlinear final RMSNorm is not folded into that patch.
- The conjunction experiment is still a separate representation/composition study; its pilot results should not be conflated with the headline reproduction.

Next validation step

The 27B Wikitext pilot fit and causal evaluation are complete under a capped Modal run. The next validation track is `tuned-pile-repro-v1`: fit on Pile validation, score held-out Pile test, then rerun causal fixtures only after the predictive checks pass. Publish only with the exact base-model revision, fit corpus, objective, dtype/quantization, hashes, and known limitations.

Sources and provenance

Research-assistance provenance

This report and its supporting scripts were developed collaboratively with **GPT-5.6 Luna Medium**, operating through OpenAI Codex in the local workspace under the researcher's direction. The model assisted with implementation, debugging, analysis, and documentation; it did not supply model outputs or replace the recorded experimental runs. Human review and reruns remain responsible for scientific claims.

Modal execution provenance

The 27B tuned lens was fit and evaluated on the persistent Modal volumes using the validated NVIDIA PyTorch 25.11 image. These are the successful app records; the links open Modal's logs and resource telemetry.

Run	Purpose	App	Observed runtime	Actual billed cost	Configured worst-case ceiling
Fit	100-step native tuned-lens fit, A100-	ap-snTfK7M83nxfrRktoZE10K	~193 s fit phase	$0.2458	$3.02

Run	Purpose	App	Observed runtime	Actual billed cost	Configured worst-case ceiling
	80GB				
Causal evaluation	27B tuned rows for verbal, two-hop, flexible	ap-rA96vrmDrqJZUPQVUUqQG7	~9 min app lifetime	$0.5071	~$1.79

The ceilings are conservative resource-request estimates from the runner configuration, including the 20-minute startup allowance; they are not a Modal invoice. The actual billed subtotal for these two successful app IDs was **$0.7528**. The full same-day workspace report was **$1.9379** across 23 billing entries, including failed/aborted attempts and shell usage. The archived billing summary is `data/provenance/modal-billing-2026-07-14.json`. The fit gate was hard-capped at $5, and the fit plus evaluation ceilings remain below that cap when considered together.

Documentation and logs

All Markdown research notes are rendered into browser-readable HTML pages alongside this report. Open the documentation index to browse the full log and experiment archive.

- Artifact storage and retrieval (ARTIFACTS.md)
- Local experiment tracking (EXPERIMENT_TRACKING.md)
- Hypothesis–Test–Evidence Traceability Matrix (HYPOTHESIS_TEST_MATRIX.md)
- Repository and fork guide (REPOSITORY_GUIDE.md)
- J-Lens Research & Experiments Log (RESEARCH_LOG.md)
- Experiment-runner operations standard (RUNNER_OPERATIONS.md)
- Tuned-lens reproduction track (TUNED_LENS_REPRODUCTION.md)
- Anthropic J-lens baseline reproduction (experiments/ANTHROPIC_BASELINE.md)
- Anthropic baseline reproduction log (experiments/ANTHROPIC_BASELINE_LOG.md)
- Conjunction swap experiment (experiments/CONJUNCTION_SWAP.md)
- Conjunction swap experiment log (experiments/CONJUNCTION_SWAP_LOG.md)
- Conjunction swap experiment: preregistration draft (experiments/CONJUNCTION_SWAP_PREREGISTRATION.md)
- Conjunction swap experiment: summary (experiments/CONJUNCTION_SWAP_SUMMARY.md)
- Encoder J-lens exploratory smoke (experiments/ENCODER_JLENS_SMOKE.md)
- Flexible generalization experiment log (experiments/FLEXIBLE_GENERALIZATION_LOG.md)
- H-lens conjunctive interaction experiment (experiments/HLENS_CONJUNCTION.md)
- H-lens conjunctive interaction experiment log (experiments/HLENS_CONJUNCTION_LOG.md)

- Two-hop causal J-lens experiment (experiments/MULTIHOP_CAUSAL.md)
- Two-hop causal J-lens experiment log (experiments/MULTIHOP_CAUSAL_LOG.md)
- Causal verbal-report experiment (experiments/VERBAL_REPORT_CAUSAL.md)
- Causal verbal-report experiment log (experiments/VERBAL_REPORT_CAUSAL_LOG.md)

Generated by `scripts/generate_research_report.py`. The report is reproducible from the JSON files under `data/experiments/`.